Mauro Julien Brochier

Technical Game Designer with a background in Computer Engineering and Computer Science, specializing in gameplay design. I focus on prototyping and implementing gameplay systems, combat, AI behaviors and player mechanics, combining design iteration with hands-on technical development. Experienced in gameplay documentation, balancing, 3D systems and adapting to proprietary development tools and scripting environments.

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Technical Game Designer	

EXPERIENCE

Game Design & Programming Instructor (RM) EDISON SCHOOL — Oct 2025 – Present

- Teach Game Design and Game Programming to students across different age groups and skill levels, from beginner to advanced.
- Deliver practical training in Unity, C# and Visual Scripting.
- Mentor students through the development of individual game projects, from initial concept to implementation.
- Taught 9 different classes to date.

Course: Game Developer – Unity 3D

Game Designer & Programmer (RM) CRISTALTEC SPA — Feb 2022 – Aug 2022

- Designed and implemented a rolling-card gameplay feature, defining the conditions, timing and behaviour governing how cards fall.
- Adapted to a proprietary programming language and development tools, applying them to gameplay development within an established technical environment.

Videogame Designer & Programmer (RM) NEXTREK SRL — Jul 2021 – Aug 2021

- Designed and implemented a complete NPC social interaction system for a 3D Unity game, inspired by The Sims, using Finite State Machines (FSM) and Behavior Trees.
- Developed a range of NPC interactions and behaviours, including conversations, flirting, dancing, emotional reactions, information exchange, short-term memory and group interactions.

University internship (RM) NEXTREK SRL — Mar 2021 – Apr 2021

- Independently conceived, designed and developed *Maze's Soul* as an individual game project during a 238-hour professional internship.
- Managed the entire development process, from initial concept and gameplay design to technical implementation.

EDUCATION

MSc in Computer Science - Videogame design and programming (MASTER'S DEGREE) Università degli Studi di MILANO — 2023 – Oct 2026

106/110 • MI

- LM-18 - 2nd level degree in Computer Science
- Thesis title: Framework Euristico per il Dynamic Difficulty Adjustment (DDA), mediante un'Architettura 'Glass-Box' a supporto del Game Designer
- Thesis subject: Videogame Design

Supervisor: Ripamonti Anna Laura

Thesis keywords: videogame design, DDA, dynamic difficulty adjustment

BSc in Computer Engineering (BACHELOR'S DEGREE) Università degli Studi 'Roma Tre' — 2015 – 2021

RM

- L-8 - 1st level degree in Information technology
- Thesis title: Player's memory, from 2d to 3d [Videogame - Maze's Soul]
- Thesis subject: Game Design & Psychology

Supervisor: DI BATTISTA GIUSEPPE

Thesis keywords: Designer, programmer, psychology

PROJECTS - VIDEOGAMES (CLICK THE TITLE TO OPEN THE PROJECT WEBSITE)

Dynamic Difficulty Adjustment Framework - Master's Thesis {Unity • C#} — Oct 2025 – Oct 2026

PERSONAL PROJECT

Designed and independently developed a research-informed heuristic Dynamic Difficulty Adjustment (DDA) framework, subsequently integrated into a Soulslike game prototype.

Personal Contributions

- Conceived and independently developed the complete DDA framework, from initial research and system design to technical implementation.
- Applied findings from player psychology and game-design research to inform the system's design.
- Integrated the framework into a Soulslike prototype and conducted a playtest with 15 players, collecting encouraging preliminary results.

A Child's Tale: Whisper of the Puppeteer {Unity · C# · FMOD Studio · Steam Audio} Apr 2024 – Jul 2024
PERSONAL PROJECT

Role: Game Designer · Gameplay Programmer · Solo Developer

First-person horror exploration game for PC and VR, built around a mysterious narrative and an audio-driven atmosphere.

Personal Contributions

- Independently conceived and developed the project's core concept, gameplay design and sound design.
- Designed and implemented layered audio systems in FMOD Studio, using crossfade transitions and auralization to reinforce atmosphere, tension and narrative progression.
- Implemented a heuristic-driven boss behaviour system, triggering appearances at specific locations throughout the house to disrupt player exploration.
- Designed and implemented interactive environments, including a fully playable piano, interactive objects and doors, and two NPCs with dialogue.
- Implemented and adapted gameplay and interaction systems for both PC and headset-based VR, ensuring a fully playable experience in each mode.

Pomegranade: Limbo {Unity · C# · Mirror} Apr 2025 – Jul 2025
Role: Team Leader · Lead Designer · Combat Designer · Game Programmer

2–4 player cooperative survival game featuring resource-management phases by day and wave-based combat by night.

Personal Contributions

- Led the game's design direction, coordinating task distribution and overseeing design decisions across the team.
- Authored the entire Technical Design Document (TDD) and approximately 70% of the Game Design Document (GDD).
- Designed and implemented the shooting and combat systems, including enemy attack mechanics and player interactions.
- Designed enemy behaviours and balancing parameters, including attack frequency, damage output and their objective of extinguishing the players' campfire.
- Designed the day-night gameplay loop, with daytime resource gathering feeding into nighttime survival encounters.
- Co-developed core multiplayer gameplay features using Mirror, collaborating with the programmer responsible for the underlying networking architecture.

Team: Mauro Brochier · L. B. · L. I.
Project Type: University of Milan academic project / Personal project

Murder Mystery Incorporated {Unity · C# · Behavior Trees} Oct 2023 – Feb 2024
Role: AI Programmer · Gameplay Designer

2D puzzle-strategy game in which players manipulate NPC interactions and environmental events to orchestrate an assassination without being discovered.

Personal Contributions

- Co-designed NPC behaviours and independently implemented the complete AI system using Behavior Trees.
- Developed a custom Behavior Tree visualization tool integrated into the Unity Editor, improving AI readability and supporting team development.
- Implemented NPC behaviours including autonomous movement, interactions, information exchange and reactions to environmental events.
- Implemented NPC traits influencing their behaviours and reactions to gameplay situations.
- Implemented one of the game's five scenarios, integrating NPC AI and gameplay logic.

Team: Mauro Brochier - M. M. - A. B. - C. C.
Project Type: University of Milan academic project

Maze's Soul {Unity · C# · Android Mobile Development} Mar 2021 – Apr 2021
PERSONAL PROJECT

Role: Game Designer · Programmer

3D third-person Android maze game designed for Google Play, built around spatial memory and navigation. Players briefly observe the entire maze from a 2D top-down perspective before seamlessly transitioning into a third-person view, where they must find the exit from memory.

Personal Contributions

- Independently conceived, designed and developed the entire game, from initial concept to implementation.
- Designed and implemented the core perspective-transition mechanic, challenging players to memorize the maze layout and locate the exit after the transition.
- Developed mobile-specific gameplay and touch controls, designing the experience specifically for smartphone interaction.

Project Type: Individual project · 238-hour professional internship at Nextrek S.r.l.

<> PROJECTS - SIMULATIONS

Real-Time Water Simulation in OpenGL with CUDA Acceleration {C++ · CUDA · OpenGL · 3D Mathematics · Physics Simulation}
Feb 2025 – Apr 2025

Role: Graphics & Simulation Programmer

Real-time grid-based fluid simulation featuring wave propagation and physically interactive objects of arbitrary shapes, rendered through OpenGL.

Personal Contributions

- Independently proposed, designed and developed the complete simulation as an academic exam project.

- Implemented a grid-based wave propagation system, supporting dynamic interactions between water and objects of arbitrary shapes.
- Developed interactive physical behaviours, including floating, sinking, dragging and shape-dependent wave generation.
- Parallelized the main simulation pipeline using CUDA, focusing on computationally intensive wave physics calculations.
- Implemented realistic water rendering in OpenGL, incorporating Fresnel-based visual effects.

Project Type: Individual academic project · University of Milan

AWARDS - 2

New Game Designer — Jul 2025
Ubisoft, Milestone, Nacon & More

1. **Pomegranade: Limbo(Team Leader)** - **3rd place**
2. **Murder Mystery Inc(AI Designer)** - **3rd place**

Showcased the video games above at the New Game Designer event (University of Milan), delivering a pitch and conducting hands-on gameplay demo sessions with industry studios.

Category: Online Game Design - Videogame Design and Programming

SKILLS - GAME DESIGN & GAMEPLAY

Gameplay & Technical Design
Advanced
Gameplay Systems, Gameplay Mechanics, Player Mechanics, Technical Implementation, Gameplay Prototyping, Design Iteration, GDD, TDD, LDD

AI, NPC & Combat Design
Intermediate
Game AI, NPC Behaviour, Finite State Machines (FSMs), Behavior Trees, Boss & Enemy AI, Combat Systems, Melee & Shooting Mechanics, Enemy Behaviour & Balancing

SKILLS - TECHNICAL DEVELOPMENT

Game Engines & Programming
Advanced
Unity, C#, Gameplay Development, Gameplay Programming, Object-Oriented Programming

Version Control & Multiplayer Networking
Advanced
Git, Mirror Networking, Version Control, Branching, Collaborative Development, Multiplayer, Client-Server Architecture

Graphics & Systems Programming
Intermediate
C++, Systems Programming, Graphics Programming

Audio Development
Advanced
FMOD Studio, Interactive Audio, Audio Integration, Steam Audio, Spatial Audio, 3D Audio

ADDITIONAL TECHNICAL SKILLS

OpenGL, Java, C, Network Architecture
Advanced
Real-Time Rendering, Graphics Programming, GPU Computing, Scripting

CUDA, Python, Spreadsheets
Intermediate
Parallel Computing, Programming, Data Analysis, Spreadsheet Formulas, Game Balancing Sheets

LANGUAGES

French, Italian
Mother Tongue

English
B2